

MarketMind AI

Enterprise Sales & Marketing Intelligence Platform

Technical Documentation

A comprehensive reference covering architecture, technology stack, agent design, database schema, API catalogue, and deployment guide.

Document Version	2.0.0
Release Date	June 06, 2026
Platform	Web — React 18 + Flask 3
Classification	Internal / Confidential
Maintainer	Engineering Team

1. EXECUTIVE OVERVIEW

What is MarketMind AI?

MarketMind AI is an enterprise-grade B2B SaaS platform that transforms how sales and marketing teams generate intelligence, qualify leads, and craft campaigns. Unlike traditional tools that return plain text, MarketMind employs a **multi-agent LangGraph pipeline** to produce structured JSON data that renders as interactive dashboards, charts, and downloadable reports.

Core Capabilities

■ Market Intelligence

Multi-agent industry analysis with SWOT, PESTEL, competitor radar charts, and market-size projections.

■ Campaign Generator

AI-crafted campaign strategies with funnel visualization, budget allocation charts, and campaign calendars.

■ Sales Pitch Engine

Personalized pitches, objection handlers, email templates, LinkedIn outreach, and meeting agendas.

■ Lead Qualification

BANT-framework scoring (0-100), conversion probability, lead temperature, and CRM-ready output.

■ Business Insights

30/60/90-day roadmaps, opportunity matrices, risk registers, and KPI dashboards.

■ Live Stock Ticker

Real-time price feed for 17 US and Indian equities, auto-refreshing every 8 seconds.

■ CRM Pipeline

Full CRUD lead management with status tracking, grading, and revenue calculation.

■ Authentication

Email/password + Google OAuth 2.0 with Flask session management.

2. TECHNOLOGY STACK

Technology Stack

MarketMind AI is built on a modern, production-tested stack optimised for AI-heavy workloads, real-time data, and developer velocity.

Frontend

Technology	Version	Purpose	License
React	18.3.x	Component-based UI framework	MIT
Vite	5.x	Build tool & HMR dev server	MIT
Tailwind CSS	3.4.x	Utility-first styling system	MIT
Recharts	2.x	Declarative chart library	MIT
Framer Motion	11.x	Animation & gesture library	MIT
Chart.js	4.x	Dashboard canvas charts	MIT
React Router v6	6.x	SPA client-side routing	MIT
React-chartjs-2	5.x	Chart.js React bindings	MIT

Backend

Technology	Version	Purpose	License
Python	3.12	Runtime language	PSF
Flask	3.x	Lightweight WSGI web framework	BSD
Flask-CORS	4.x	Cross-Origin Resource Sharing	MIT
SQLAlchemy	2.x	ORM for PostgreSQL / SQLite	MIT
psycopg2-binary	2.9	PostgreSQL adapter for Python	LGPL
python-dotenv	1.x	Environment variable loader	BSD
Gunicorn	22.x	Production WSGI HTTP server	MIT

Artificial Intelligence & Orchestration

Technology	Version	Purpose	License
Groq API	Cloud	LLM inference (ultra-fast Llama 3)	Commercial
Llama 3.3 70B	Groq	Primary language model	Meta Llama
LangChain	0.2.x	LLM chain composition layer	MIT
LangGraph	0.1.x	Multi-agent stateful graph engine	MIT
LangChain-Groq	0.1.x	Groq provider for LangChain	MIT
python-docx	1.1	DOCX report generation	MIT
python-pptx	0.6	PPTX report generation	MIT
ReportLab	4.x	PDF generation	BSD
Markdown	3.x	Markdown→HTML conversion	BSD

Database & Storage

Technology	Version	Purpose	License
PostgreSQL	16	Primary relational database	PostgreSQL
SQLite	3.x	Local development / fallback DB	Public Domain
cachetools	5.x	In-memory LRU cache (Redis-free)	MIT

Authentication & Security

Technology	Purpose	Standard
Flask Sessions	Server-side session management	HTTP Cookie
Google OAuth 2.0	Social login via Google account	RFC 6749
oauthlib	OAuth 2.0 client implementation	BSD
requests-oauthlib	HTTP OAuth wrapper	ISC
Werkzeug	Password hashing & security utils	BSD

3. SYSTEM ARCHITECTURE

System Architecture

High-Level Architecture

MarketMind AI follows a clean **3-tier architecture**: React SPA (Presentation) ↔ Flask REST API (Application) ↔ PostgreSQL (Data). The AI layer is embedded in the Application tier as LangGraph agent graphs that run synchronously within the Flask request/response cycle.

Tier	Technology	Responsibility
Presentation	React 18 + Vite	SPA, routing, state, charts, forms
API Gateway	Flask 3 + Flask-CORS	REST endpoints, auth, session mgmt
AI Engine	LangGraph + Groq	Multi-agent graphs, structured output
Data	PostgreSQL 16	Users, leads, history, reports, stocks
Export	ReportLab / python-docx / python-pptx	PDF / DOCX / PPTX generation
Cache	cachetools LRU	In-process response cache

Multi-Agent Pipeline

Each analysis module runs a dedicated LangGraph StateGraph. Nodes communicate through a typed state dict. Every node calls Groq with a structured JSON prompt and a schema example, then validates + merges the parsed output into the shared state.

Agent Graph	Nodes	Output
MarketIntelligenceGraph	research → competitor → synthesis → score	Structured market report JSON
CampaignStrategyGraph	audience → funnel → content → metrics	Campaign plan + KPI JSON
SalesGraph	persona → pitch → objection → email	Full sales package JSON
LeadQualificationGraph	bant → scoring → intent → action	BANT scores + lead priority JSON
BusinessConsultantGraph	challenge → opportunity → risk → roadmap	90-day plan + strategy JSON
ReportGenerationAgent	compile → template → render → export	PDF / DOCX / PPTX bytes

Request / Response Data Flow

- 1. User submits form on React page (POST /api/v2/[module])
- 2. Flask validates input, starts LangGraph graph execution
- 3. Node 1: Industry / persona research prompt → Groq → parsed JSON
- 4. Node 2: Competitive / strategy analysis → Groq → parsed JSON merged
- 5. Node 3: Synthesis + confidence scoring
- 6. Node 4: Final schema validation + fallback defaults applied
- 7. Flask returns { success: true, data: { ...structured_json } }
- 8. React renders sections: charts, tables, cards, export buttons
- 9. User clicks Export → POST /api/v2/export/pdf → binary PDF download

4. DATABASE SCHEMA

Database Schema — PostgreSQL 16

users

Column	Type	Constraints	Description
email	VARCHAR(255)	PRIMARY KEY	User email (unique identifier)
password	TEXT	NOT NULL	Hashed password
name	VARCHAR(200)	NOT NULL	Full display name
first_name	VARCHAR(100)		Given name
last_name	VARCHAR(100)		Family name
avatar	VARCHAR(10)		Initial(s) for avatar
joined_at	TIMESTAMPTZ	NOT NULL	Account creation timestamp

history

Column	Type	Constraints	Description
id	SERIAL	PRIMARY KEY	Auto-increment ID
user_email	VARCHAR(255)	FK → users	Owner
type	VARCHAR(50)	NOT NULL	campaign / pitch / lead / market / insights
title	TEXT	NOT NULL	Human-readable label
input_data	JSONB	NOT NULL	Form input snapshot
result	TEXT	NOT NULL	Raw v1 text result
created_at	TIMESTAMPTZ	NOT NULL	Creation time

crm_leads

Column	Type	Constraints	Description
id	SERIAL	PRIMARY KEY	
user_email	VARCHAR(255)	FK → users	Owner
name	VARCHAR(200)	NOT NULL	Lead / contact name

Column	Type	Constraints	Description
company	VARCHAR(200)		Organisation
score	INTEGER	CHECK 0-100	Lead quality score
grade	CHAR(1)		A / B / C / D
status	VARCHAR(50)	NOT NULL	New / Contacted / Proposal / Closed Won
details	TEXT		Free-text notes
created_at	TIMESTAMPTZ	NOT NULL	

stock_prices

Column	Type	Constraints	Description
ticker	VARCHAR(20)	PRIMARY KEY	Ticker symbol
name	VARCHAR(200)	NOT NULL	Full security name
price	NUMERIC(14,2)	NOT NULL	Latest price
change_pct	NUMERIC(6,2)		Intraday % change
history	JSONB	NOT NULL	12-month price array
last_updated	TIMESTAMPTZ	NOT NULL	

NEW TABLE — analysis_reports

Column	Type	Constraints	Description
id	SERIAL	PRIMARY KEY	
user_email	VARCHAR(255)	FK → users	Owner
module	VARCHAR(50)	NOT NULL	market / campaign / pitch / lead / insights
title	TEXT	NOT NULL	
input_data	JSONB	NOT NULL	Form inputs
result_json	JSONB	NOT NULL	Full structured agent output
confidence_score	SMALLINT		AI confidence 0-100
created_at	TIMESTAMPTZ	NOT NULL	

NEW TABLE — exported_reports

Column	Type	Constraints	Description
id	SERIAL	PRIMARY KEY	
user_email	VARCHAR(255)	FK → users	Owner
report_id	INTEGER	FK → analysis_reports	Parent analysis
format	VARCHAR(10)	NOT NULL	pdf / docx / pptx
file_path	TEXT	NOT NULL	Server-side file path
file_size	INTEGER		Bytes
created_at	TIMESTAMPTZ	NOT NULL	

NEW TABLE — watchlists

Column	Type	Constraints	Description
id	SERIAL	PRIMARY KEY	
user_email	VARCHAR(255)	FK → users	Owner
industry	VARCHAR(200)	NOT NULL	Industry / topic label
keywords	JSONB	NOT NULL	Array of monitored keywords
created_at	TIMESTAMPTZ	NOT NULL	

5. API ENDPOINT CATALOGUE

API Endpoint Catalogue

All endpoints require an active Flask session cookie unless marked (public). v1 endpoints return raw markdown text. v2 endpoints return structured JSON.

Authentication Endpoints

Method	Path	Description
POST	/api/auth/login	Email + password login, sets session cookie
POST	/api/auth/register	Create new user account
POST	/api/auth/logout	Clear session cookie
GET	/api/auth/me	Return current session user data
GET	/api/auth/google	Initiate Google OAuth 2.0 flow
GET	/api/auth/google/callback	Handle OAuth redirect + set session
POST	/api/auth/reset-password	Trigger password reset email

v1 AI Generation Endpoints (Legacy — Text Output)

Method	Path	Description
POST	/api/market_analysis	Raw markdown market analysis text
POST	/api/generate_campaign	Raw markdown campaign strategy text
POST	/api/generate_pitch	Raw markdown sales pitch text
POST	/api/lead_score	Raw markdown lead scoring text
POST	/api/business_insights	Raw markdown business insights text
POST	/api/translate	Translate result text to target language
POST	/api/refine_result	Refine/edit result with follow-up prompt
POST	/api/export_pdf	Export v1 text result as PDF

v2 Structured AI Endpoints (Multi-Agent JSON Output)

Method	Path	Returns
POST	/api/v2/market_analysis	Full market intel JSON (SWOT, PESTEL, charts data)
POST	/api/v2/campaign	Campaign plan JSON (funnel, KPIs, calendar)
POST	/api/v2/pitch	Sales package JSON (pitch, emails, objections)
POST	/api/v2/lead_score	BANT JSON (score, probability, actions)
POST	/api/v2/business_insights	Strategy JSON (30/60/90 plan, matrix)
POST	/api/v2/export/pdf	Binary PDF bytes (Content-Disposition: attachment)
POST	/api/v2/export/docx	Binary DOCX bytes
GET	/api/v2/reports	List user's analysis_reports (paginated)
DELETE	/api/v2/reports/:id	Delete a saved report
GET	/api/v2/analytics/summary	Aggregated usage stats for Analytics dashboard
POST	/api/v2/watchlist	Add industry to user's watchlist
GET	/api/v2/watchlist	Get user's watchlists

CRM & Data Endpoints

Method	Path	Description
GET	/api/crm/leads	List user's CRM leads
POST	/api/crm/leads	Add new lead to CRM
PATCH	/api/crm/leads/:id	Update lead status
DELETE	/api/crm/leads/:id	Remove lead from CRM
GET	/api/history	Get user's generation history
POST	/api/history/save	Manually save a result to history
DELETE	/api/history/:id	Delete a history record
GET	/api/dashboard/stats	Live dashboard KPI + stock prices

6. FRONTEND STRUCTURE

Frontend Structure & Pages

Page Routes

Route	Component	Auth	Description
/	Dashboard.jsx	✓	Live KPI cards, stock ticker, charts
/market-analysis	MarketAnalysis.jsx	✓	Multi-agent market intelligence
/campaign	CampaignGenerator.jsx	✓	AI campaign strategy builder
/pitch	SalesPitch.jsx	✓	Sales pitch & outreach tools
/lead-scoring	LeadScoring.jsx	✓	BANT lead qualification engine
/business-insights	BusinessInsights.jsx	✓	Strategic intelligence & roadmaps
/analytics	Analytics.jsx	✓	Aggregated platform analytics (NEW)
/reports	ReportHistory.jsx	✓	Report history & downloads (NEW)
/crm	CRM.jsx	✓	CRM lead pipeline management
/history	History.jsx	✓	Generation history log
/profile	Profile.jsx	✓	User account settings
/themes-store	ThemesStore.jsx	✓	UI theme customization
/login	Login.jsx	Public	Email + Google OAuth login
/register	Register.jsx	Public	New account registration
/reset-password	ResetPassword.jsx	Public	Password reset flow

Chart Component Library (components/charts/)

Component	Chart Type	Used In
MarketGrowthChart	LineChart	Market Analysis — growth projections
CompetitorRadar	RadarChart	Market Analysis — competitor scoring
MarketSharePie	PieChart	Market Analysis — market share
OpportunityHeatmap	ScatterChart	Market Analysis — effort vs impact
ConversionFunnel	BarChart	Campaign — funnel stages

Component	Chart Type	Used In
BudgetAllocation	BarChart	Campaign — channel spend split
LeadGauge	Custom SVG	Lead Scoring — 0-100 arc gauge
BANTScoreBar	Custom SVG	Lead Scoring — BANT factors
AgentProgressBar	Custom CSS	All v2 pages — agent node steps
SkeletonCard	CSS pulse	All v2 pages — loading placeholders

7. PROJECT FOLDER STRUCTURE

Project Folder Structure

```

resilient-pascal/ ■■■ app.py ← Flask application entry point ■■■ database.py ← DB connection,
models, helpers ■■■ generate_docs.py ← Documentation PDF generator ■■■ requirements.txt ← Python
package manifest ■■■ .env ← Secrets (gitignored) ■■■ .gitignore ■■■ agents/ ← Multi-agent
LangGraph graphs ■ ■■■ __init__.py ■ ■■■ base.py ← Shared chain, JSON parser, retry ■ ■■■
market_intelligence.py ■ ■■■ campaign_strategy.py ■ ■■■ sales_agent.py ■ ■■■
lead_qualification.py ■ ■■■ business_consultant.py ■ ■■■ report_generation.py ■■■ exports/ ←
Generated reports (gitignored) ■■■ static/ ← Flask static assets ■■■ templates/ ← Flask HTML
templates ■■■ frontend/ ■■■ index.html ■■■ package.json ■■■ vite.config.js ■■■ src/ ■■■
App.jsx ■■■ main.jsx ■■■ index.css ■■■ pages/ ■ ■■■ Dashboard.jsx ■ ■■■ MarketAnalysis.jsx ■
■■■ CampaignGenerator.jsx ■ ■■■ SalesPitch.jsx ■ ■■■ LeadScoring.jsx ■ ■■■
BusinessInsights.jsx ■ ■■■ Analytics.jsx ← NEW ■ ■■■ ReportHistory.jsx ← NEW ■ ■■■ CRM.jsx ■
■■■ History.jsx ■ ■■■ Profile.jsx ■ ■■■ ThemesStore.jsx ■ ■■■ Login.jsx ■ ■■■ Register.jsx ■
■■■ ResetPassword.jsx ■■■ components/ ■ ■■■ Navbar.jsx ■ ■■■ ChatPanel.jsx ■ ■■■
TranslatorBlock.jsx ■ ■■■ charts/ ← NEW ■ ■■■ MarketGrowthChart.jsx ■ ■■■ CompetitorRadar.jsx
■ ■■■ MarketSharePie.jsx ■ ■■■ OpportunityHeatmap.jsx ■ ■■■ ConversionFunnel.jsx ■ ■■■
BudgetAllocation.jsx ■ ■■■ LeadGauge.jsx ■ ■■■ BANTScoreBar.jsx ■ ■■■ AgentProgressBar.jsx ■
■■■ SkeletonCard.jsx ■■■ context/ ■ ■■■ AuthContext.jsx ■ ■■■ ThemeContext.jsx ■■■ services/
■■■ api.js ← v1 API calls ■■■ apiV2.js ← NEW – v2 structured + export calls

```

8. ENVIRONMENT & DEPLOYMENT

Environment Variables & Deployment

Required Environment Variables (.env)

Variable	Example Value	Description
GROQ_API_KEY	gsk_xxx...	Groq Cloud API key for LLM inference
SECRET_KEY	random-32-char-string	Flask session signing key
DATABASE_URL	postgresql://user:pw@host/db	PostgreSQL connection string
GOOGLE_CLIENT_ID	xxx.apps.googleusercontent.com	Google OAuth client ID
GOOGLE_CLIENT_SECRET	GOCSPX-xxx	Google OAuth client secret

Local Development Setup

- 1. Clone repository and cd into project root
- 2. Copy .env.example to .env and fill in all variables
- 3. Install Python deps: pip install -r requirements.txt
- 4. Start PostgreSQL and create database: createdb marketmind
- 5. Run backend: python app.py (starts on port 5000)
- 6. Install frontend deps: cd frontend && npm install
- 7. Start frontend: npm run dev (starts on port 5173)
- 8. Open browser at http://localhost:5173

Implementation Roadmap

Sprint	Scope	Duration	Status
Sprint 1	Backend: agents/ + v2 endpoints + PostgreSQL migration	3 days	In Progress
Sprint 2	Market Analysis v2 output + chart library	2 days	Planned
Sprint 3	Campaign, Pitch, Lead, Insights v2 output	3 days	Planned
Sprint 4	Analytics page, Reports page, Polish	2 days	Planned